

Cover Letter

Original submission to Wireless Personal Communications

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To: The Editor-in-Chief, *Wireless Personal Communications* (Springer)

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Manuscript title: A Lightweight Complex-Valued CNN with Complex Squeeze-and-Excitation Attention for Single-Channel Blind Source Separation of Co-Frequency Communication Signals

Manuscript type: Original Research Paper

Dear Editor,

We are pleased to submit the above manuscript for consideration as an original research article in *Wireless Personal Communications*. The paper proposes a lightweight complex-valued CNN with a Complex Squeeze-and-Excitation (C-SE) attention block for single-channel blind source separation of co-frequency communication signals, and evaluates it against both convolutional and state-space baselines. We believe the topic fits the journal's scope in wireless communications, signal processing, and interference mitigation.

Key strengths of the manuscript.

1. **Novel complex-native attention.** The C-SE block jointly pools real and imaginary statistics and produces a *real-valued* channel weight that scales complex features while preserving their phase—to the best of our knowledge, a phase-preserving, real-gated design not previously evaluated for lightweight single-channel blind separation of near-co-frequency communication signals.
2. **Lightweight and deployable.** The full model has only 235K parameters and trains in minutes on a single 8 GB GPU (NVIDIA RTX 4060), making it suitable for resource-constrained edge devices and software-defined radios.
3. **Rigorous multi-seed evaluation.** Five random seeds with sample standard deviations are reported for the proposed model and the parameter-matched baselines, together with paired Wilcoxon signed-rank tests on per-seed SDR. We are explicit that no comparison reaches statistical significance at $\alpha = 0.05$ and position the model honestly on per-parameter efficiency and partial collapse-resistance rather than claiming raw SDR dominance.
4. **Comprehensive baselines and ablations.** Four modulation types (BPSK, QPSK, 8PSK, 16QAM), a complex-domain Conv-TasNet, parameter-matched real-valued and no-SE ablations, and scaled-down re-implementations of two cross-domain communication-signal baselines (CNSE, S4-UNET); ablations isolate the SE block, the scale mode, and the squeeze pooling.
5. **Generalisation to the co-frequency limit.** A model retrained on carrier-gap distributions $U(0, 5 \text{ Hz})$ holds $\text{SDR} \approx 3.3 \text{ dB}$ across $[0, 500 \text{ Hz}]$, including the previously unseen co-frequency limit $\Delta f = 0 \text{ Hz}$.
6. **Reproducibility.** The training, evaluation, and figure-generation code is publicly available at https://github.com/BinNong/single_channel_source_seperation under the

MIT license; per-seed numbers, exact commands, and hyperparameters are recorded in docs/EXPERIMENT_LOG.md, and pre-trained checkpoints will be made publicly available upon publication.

Headline numbers (SNR = 10 dB). The proposed C-SE reaches SDR = 2.31(62) dB at 235K parameters (five seeds). The matched no-SE complex CNN reaches 1.81(48) dB (five seeds); the matched real-valued CNN reaches 2.20(59) dB (five seeds); the complex Conv-TasNet reaches 3.23(40) dB (three seeds, 817K parameters). CNSE (6.69M parameters) reaches 2.38(91) dB and S4-UNET (1.57M) reaches 3.05 dB (three seeds each). C-SE is therefore competitive at the 235K-parameter scale on per-parameter efficiency, partial collapse-resistance, and predictable runtime.

Originality and ethics.

- The manuscript is original, has not been published elsewhere, and is not under consideration by any other journal.
- All authors have approved the submission; the authors have no conflicts of interest to declare.
- No human subjects or animal experiments are involved; all experiments use synthetic signals generated by our open-source data generator.

Author contributions. B. Nong conceived the C-SE design, ran the multi-seed experiments, and drafted the manuscript. W. Fu contributed the matched-baseline analysis and the scale-mode / pooling ablations. Z. Jiang contributed the cross-domain baseline re-implementations (CNSE, S4-UNET) and the statistical-significance analysis. All authors reviewed and approved the final manuscript.

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Suggested reviewers. We have no specific reviewer suggestions; we trust the editorial board to identify expert reviewers in signal processing for wireless communications.

Thank you for considering our manuscript. We look forward to hearing from you.

Sincerely,

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